Merging Reconstructions with Landmark Schur Complements

Joint camera poses and intrinsics across local coordinate frames

Abstract

Independently reconstructed image clusters can be merged by summarizing each cluster as a joint likelihood over its cameras. We linearize the local reprojection problem and eliminate only landmark increments, retaining camera poses, intrinsics, and their correlations. Each resulting Gaussian system is factored into a Bayes tree and expressed in a common frame through a leaf-specific similarity transformation. Shared images identify common pose and calibration variables, and one root optimization reconciles all leaf evidence. This note derives the Schur reduction, coordinate constraints, similarity-aware residuals, analytic derivatives, and final camera information, and states the assumptions under which the frozen approximation is useful.

1 Local reconstructions and shared camera variables

The input is a hierarchy of reconstructed image clusters. Each leaf supplies camera poses, intrinsics, image measurements, and 3D landmarks in its own coordinate frame. Overlapping leaves can contain the same image but assign it different local indices and different estimates. Image filenames establish identity across leaves. The merged state contains one global pose and calibration per image and one similarity transformation per leaf.

Symbol	Meaning
ℓ, i, j	Leaf, globally identified image, and leaf-local landmark indices.
$\bar{T}_{\ell i} \in \mathrm{SE}(3)$	Frozen camera-to-leaf pose estimate.
$\bar{\kappa}_{\ell i} \in \mathbb{R}^3$	Frozen `Cal3Bundler` parameters (f, k_1, k_2).
$\bar{p}_{\ell j} \in \mathbb{R}^3$	Frozen landmark position in the leaf frame.
T_i, κ_i	Global camera pose and per-image calibration to optimize.
$S_\ell \in \mathrm{Sim}(3)$	Similarity mapping global coordinates into leaf ℓ.

The observation model uses one focal length and two radial-distortion coefficients. For $q = T^{-1}p$, let $x = q_x/q_z$, $y = q_y/q_z$, and $r^2 = x^2 + y^2$. Then

$$\pi(T, \kappa, p) = \begin{bmatrix} f(1 + k_1 r^2 + k_2 r^4)x + u_0 \\ f(1 + k_1 r^2 + k_2 r^4)y + v_0 \end{bmatrix}. \tag{1}$$

The principal point (u_0, v_0) is fixed. Occurrences of a shared image must agree on it. Different images have independent calibrations, while occurrences of the same image share one global calibration.

Each leaf's reprojection model is linearized once at its supplied estimates. The reconstruction-based model uses isotropic pixel noise $\sigma^2 I_2$, with $\sigma = 1$ by default, and excludes tracks observed by fewer than two distinct retained cameras. The resulting camera likelihood retains the weights of the linearized model. After landmark elimination, merging uses only that likelihood, the frozen camera estimates, and image identities; it does not revisit the points.

2 Eliminating landmarks while retaining intrinsics

For a leaf, collect pose increments $\delta x \in \mathbb{R}^{6n}$, calibration increments $\delta\kappa \in \mathbb{R}^{3n}$, and landmark increments $\delta p \in \mathbb{R}^{3m}$. Whitened linearization gives the local least-squares problem

$$\min_{\delta x,\delta\kappa,\delta p} \frac{1}{2} \|A_x\delta x + A_\kappa\delta\kappa + A_p\delta p - b\|^2, \qquad b = -r(\bar{T},\bar{\kappa},\bar{p}). \tag{2}$$

The residual offset b is retained even when the supplied reconstruction is not at a stationary point. Set $\delta c = [\delta x^\top \ \delta\kappa^\top]^\top$ and $A_c = [A_x \ A_\kappa]$. The normal equations have block structure

$$\begin{bmatrix} H_{cc} & H_{cp} \\ H_{pc} & H_{pp} \end{bmatrix} \begin{bmatrix} \delta c \\ \delta p \end{bmatrix} = \begin{bmatrix} g_c \\ g_p \end{bmatrix}, \qquad H = A^\top A, \quad g = A^\top b. \tag{3}$$

If H_{pp} is nonsingular, profiling out the points yields

$$H_{\text{red}} = H_{cc} - H_{cp}H_{pp}^{-1}H_{pc}, \tag{4}$$

$$g_{\text{red}} = g_c - H_{cp}H_{pp}^{-1}g_p. \tag{5}$$

The pose–intrinsics block is therefore

$$(H_{\text{red}})_{x\kappa} = H_{x\kappa} - H_{xp}H_{pp}^{-1}H_{p\kappa}. \tag{6}$$

Preserving this block allows calibration changes to influence the pose solution and preserves correlations between images induced by shared landmarks.

2.1 Square-root reduction

The implementation uses QR elimination rather than explicitly forming Equation (4). Conceptually, an orthogonal transformation gives

$$Q^\top \begin{bmatrix} A_p & A_c & b \end{bmatrix} = \begin{bmatrix} R_p & B_p & d_p \\ 0 & B & d \end{bmatrix}. \tag{7}$$

When the landmark columns are eliminable, solving the first block for δp leaves the reduced cost

$$E_{\text{red}}(\delta c) = \tfrac{1}{2}\|B\delta c - d\|^2, \qquad B^\top B = H_{\text{red}}, \quad B^\top d = g_{\text{red}}. \tag{8}$$

Any additive constant discarded by the factor representation does not affect optimization. GTSAM's `GaussianFactorGraph.marginal(retained_keys, EliminateQR)` retains the pose and calibration keys and eliminates only landmark keys. The returned producer summary is ungauged.

2.2 The retained camera likelihood

The reduced state has $9n$ coordinates: six pose and three calibration coordinates per image. The retained likelihood includes pose–pose, calibration–calibration, and pose–calibration interactions. Replacing it with independent priors on each camera would generally lose information about how cameras can change together. The linear term is equally essential: g_{red} determines the preferred increment when the observations have nonzero residuals. An information matrix alone is insufficient, since x^2 and $(x-3)^2$ have the same quadratic coefficient but different optima. The summary therefore preserves both B and d, or an equivalent collection of Gaussian factors, together with its linearization point.

3 Gauge constraints and leaf Bayes trees

An observable monocular leaf has seven similarity gauge freedoms: three rotations, three translations, and one scale. Intrinsics remain in the state, so for n cameras the expected ungauged rank is $9n - 7$. Additional self-calibration or geometric ambiguities are rejected; the code does not add implicit calibration priors to make such a system invertible.

The consumer fixes a local coordinate chart before factoring an ungauged leaf. It adds six exact rows fixing the first pose at its frozen estimate and one exact row fixing the baseline length to the farthest camera, linearized at those estimates. A nonzero baseline is required. These constraints act only on poses, not on calibration variables. GTSAM's `NonlinearEqualityPose3` and constrained `RangeFactorPose3` supply the rows.

The augmented graph is factored by multifrontal QR with GTSAM's default ordering. Its Gaussian Bayes-tree conditionals jointly represent the constrained camera quadratic. A conditional can contain any mixture of pose and calibration blocks; block widths are six and three, respectively. Conditionals preserve their linear terms and noise models, including exact constraints.

A supplied camera summary may already include coordinate constraints. In that case its retained system must be full rank, and no second gauge is added. Gauged and ungauged leaves can coexist in the hierarchy. Rank checks normalize Jacobian columns and tolerate numerical perturbations of gauge null modes. The constraints must express a choice of coordinates rather than unintended geometric priors; rank alone cannot distinguish these meanings.

4 Lifting the leaf quadratic to global variables

A similarity variable lets each leaf's constrained local chart move within the global reconstruction. Let $\Phi(S, T)$ denote GTSAM's `Similarity3.transformFrom(Pose3)`. With GTSAM's stored parameters $S = (R_s, t_s, s)$, its action is

$$p \longmapsto s(R_s p + t_s), \qquad \Phi(S, (R, t)) = (R_s R,\ s(R_s t + t_s)). \tag{9}$$

The translation parameter is inside the scale in this convention. The code uses the library operation and its derivatives, rather than implementing an alternate similarity convention.

For each image in leaf ℓ, define frozen-chart increments

$$\xi_{\ell i}(T_i, S_\ell) = \mathrm{Local}_{\bar{T}_{\ell i}}(\Phi(S_\ell, T_i)) \in \mathbb{R}^6, \tag{10}$$

$$\eta_{\ell i}(\kappa_i) = \mathrm{Local}_{\bar{\kappa}_{\ell i}}(\kappa_i) = \kappa_i - \bar{\kappa}_{\ell i} \in \mathbb{R}^3. \tag{11}$$

For conditional q, concatenate the appropriate increments in its actual key order into $z_{\ell q}$. The lifted residual is

$$e_{\ell q} = A_{\ell q} z_{\ell q} - b_{\ell q}. \tag{12}$$

Each shared image has one global calibration but may have different frozen calibration estimates in different leaves. Similarities affect poses only: neither focal length nor distortion coefficients change under a change of world frame or world scale. The residual is passed unwhitened to a `CustomFactor` with the conditional's original noise model, so weighting is applied exactly once.

5 Derivatives, root optimization, and the final reduction

The lifted factors use analytic derivatives in GTSAM's local retraction charts. For a pose block, let A_i^x be its six columns in a conditional and define

$$J_i^{\mathrm{loc}} = D_2 \operatorname{Local}_{\bar{T}_{\ell i}}(\Phi(S_\ell, T_i)), \quad J_i^S = D_S \Phi(S_\ell, T_i), \quad J_i^T = D_T \Phi(S_\ell, T_i). \tag{13}$$

The chain rule gives

$$\frac{\partial e_{\ell q}}{\partial \delta T_i} = A_i^x J_i^{\mathrm{loc}} J_i^T, \tag{14}$$

$$\frac{\partial e_{\ell q}}{\partial \delta S_\ell} = \sum_{i \text{ in pose blocks}} A_i^x J_i^{\mathrm{loc}} J_i^S, \tag{15}$$

$$\frac{\partial e_{\ell q}}{\partial \delta \kappa_i} = A_i^\kappa. \tag{16}$$

Calibration contributes no similarity derivative. In particular, a calibration-only conditional has an identically zero similarity Jacobian. The implementation computes column offsets from the variable types; it never assumes that every conditional block has six columns. These formulas apply at nonidentity poses and nonunit similarity scales, not just at initialization.

5.1 Initialization and one global solve

Camera identity is resolved by filename, and sorted filenames determine the output indices. Pose-based similarity alignment initializes a connected graph of overlapping leaves. Each usable overlap link needs at least two shared cameras with distinct centers. Global calibrations are initialized from the first occurrence of each image in leaf traversal order.

The input scene tree assembles the leaf evidence bottom-up, without performing an optimization at internal nodes. The root problem is

$$\min_{\{T_i, \kappa_i, S_\ell\}} \sum_{\ell, q} \tfrac{1}{2} \| e_{\ell q}(T, \kappa, S_\ell) \|^2_{\Sigma_{\ell q}}, \qquad S_0 = I, \tag{17}$$

Here $\|e\|^2_\Sigma = e^\top \Sigma^{-1} e$ for stochastic rows; zero-noise rows are exact constraints. Fixing the first leaf's similarity selects the global frame and scale. Levenberg–Marquardt with multifrontal QR solves this problem once at the root. The local pose and calibration estimates used inside the factors remain frozen throughout optimization. The public leaf-to-global transform is S_ℓ^{-1}.

5.2 Final camera information retains calibration

At the converged solution, the assembled graph is linearized again with respect to its global camera and similarity variables. This relinearizes the lifted factors, not the original image measurements. The final marginalization removes only the similarity increments:

$$E_{\mathrm{final}}(\delta c) = \min_{\delta s} E_{\mathrm{root,lin}}(\delta c, \delta s), \qquad \delta c = (\delta x, \delta \kappa). \tag{18}$$

The resulting Gaussian Bayes tree contains global `X(i)` pose keys and `K(i)` calibration keys. It contains no landmarks or similarities and includes the chosen coordinate gauge. The returned scene has optimized poses and calibrations, image metadata, and zero tracks. Its quadratic describes local information in the frozen approximation, not the uncertainty of a newly optimized full bundle-adjustment problem.

6 The complete merging procedure

The algorithm separates measurement reduction from camera merging. A leaf summary contains the reduced Gaussian factors, frozen pose and calibration estimates, image identities, local key associations, and a declaration of whether its coordinate gauge is fixed. The global optimizer needs no landmark positions or image observations once these summaries have been constructed.

1. **Linearize each leaf.** Assemble the weighted local observation model at the supplied poses, intrinsics, and points, retaining the residual offsets.
2. **Eliminate landmarks.** Apply QR to the point columns and retain all pose and calibration columns. Preserve the joint quadratic and linear term.
3. **Choose local coordinates and factor.** For each ungauged leaf, fix one pose and one baseline length. Factor the retained graph into Gaussian Bayes-tree conditionals.
4. **Match images and initialize frames.** Assign one global pose and calibration to each filename. Align overlapping leaves to initialize their similarities, and initialize calibration from the first occurrence of each image.
5. **Lift and assemble.** Evaluate each conditional through its leaf's similarity and frozen camera charts. Collect the leaf factors bottom-up, counting each leaf's contribution once.
6. **Optimize globally.** Fix the first similarity to identity and jointly refine all global poses, intrinsics, and remaining similarities at the root.
7. **Return cameras and local information.** Export the optimized cameras and leaf-to-global similarities. Linearize the merged graph and remove the similarity increments to obtain the final joint camera Bayes tree.

6.1 The scene tree and the Bayes trees have different roles

The scene tree organizes which reconstructions contribute evidence and how their summaries are collected. Internal nodes concatenate child factor graphs; they do not optimize or discard additional camera variables. Each leaf's Bayes tree is an algebraic factorization of that leaf's reduced likelihood, and the final Bayes tree factors the merged camera information. Their clique structure follows variable connectivity and elimination ordering, not the scene hierarchy.

6.2 Assumptions and approximation limits

The usable overlap graph must be connected, with at least two shared cameras at distinct centers on each link used for initialization. Each leaf must have the declared geometric rank; additional self-calibration ambiguity is not repaired by implicit calibration priors. Image identities and fixed principal points must agree across shared images. Evidence must be assigned consistently across leaves: sharing camera variables is intentional, but counting the same observation twice would duplicate its information.

Landmark elimination is exact for the linearized problem, up to additive constants. During merging, pose-chart and similarity derivatives change but the leaf quadratics remain fixed. Large departures from their linearization points can reduce accuracy. Recovering landmarks requires a subsequent step using the original observations. Uncertainty interpretation must respect the chosen gauge and the frozen-linearization approximation.